

Decoding Attention in Children with Attention Deficit Hyperactivity Disorder through Multimodal Analysis for Digital Learning

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Abstract

Digital learning platforms offer flexibility and customization but place significant demands on attention and self-regulation, challenges that are especially pronounced for children with Attention Deficit Hyperactivity Disorder (ADHD). This research addresses these challenges by developing a multimodal framework to analyze attention patterns specific to ADHD learners. By integrating physiological signals such as heart rate variability and galvanic skin response, along with behavioural cues like task engagement metrics, the study assesses attention dynamics in real-time. The goal is to identify ADHD-specific attention fluctuations and design adaptive learning interfaces to offer targeted interventions when the attention is lost. These interventions aim to optimize content delivery and sustain attention, ultimately enhancing learning outcomes for children with ADHD. The research bridges cognitive science and educational technology, offering valuable insights into how digital learning environments can be tailored to support the needs of ADHD students while advancing the broader field of attention dynamics in education.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction devices**.

Keywords

ADHD, Digital Learning Platforms, Multimodal Analysis, Adaptive Learning Interfaces, Educational Technology

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1 Introduction

While digital platforms provide flexibility and customization, they also impose considerable demands on self-regulation and sustained attention—skills that are challenging for all learners, but particularly so for children with Attention Deficit Hyperactivity Disorder (ADHD) [15, 18]. ADHD is a common neurodevelopmental disorder impacting around 5% of children in school globally [5, 28], marked by symptoms such as inattention, hyperactivity, impulsivity, and difficulties in controlling behaviour [4, 6, 13, 20]. These symptoms significantly hinder academic performance, leading to poor grades, grade retention, and long-term educational and occupational challenges for affected children [9, 10]. For children with ADHD, traditional educational settings often exacerbate these issues, making it difficult to engage and succeed in structured classroom environments.

The shift to digital learning during the COVID-19 pandemic has both expanded access to education and intensified these challenges. Digital platforms offer flexibility but also present challenges that demand sustained attention, area where children with ADHD often faces a lot of difficulties [15, 17, 22, 26]. These platforms, though customizable, often present unique obstacles for ADHD learners, including an increased likelihood of distraction and difficulty maintaining focus for prolonged periods. As such, ADHD students may find it harder to navigate the complexities of digital learning environments compared to their peers.

Despite these challenges, digital platforms hold promise for children with ADHD, as they allow for personalized pacing and engagement, potentially alleviating some difficulties experienced in traditional classrooms. However, the persistence of attention-related issues remains a major barrier to effective cognitive engagement in digital learning environments [9]. Without addressing these attention difficulties, the potential benefits of digital learning for ADHD students cannot be fully realized, underscoring the need for adaptive learning systems that can support sustained cognitive engagement in real-time.

Moreover, the availability of special educators, who play a crucial role in providing individualized support for children with ADHD, is often insufficient, particularly in developing countries like India [9]. This lack of resources emphasizes the need for adaptive digital learning systems that can provide real-time support to students with ADHD, reducing the dependency on constant human intervention. In this context, understanding how children with ADHD engage with digital platforms, especially in terms of attention allocation, is

crucial to developing systems that can accommodate their unique learning needs.

Traditional methods of diagnosing and monitoring ADHD are often limited by their subjective nature, relying heavily on reports from parents, teachers, or clinicians, which can introduce inaccuracies and delays in identifying attentional difficulties. Consequently, there is a pressing need to develop innovative programs that utilize Artificial Intelligence (AI) to objectively assess and track attention challenges in children with ADHD. Such systems can bridge the gap in monitoring and provide timely and effective support.

My PhD research aims to enhance learning outcomes for children with ADHD in digital environments by analyzing their attention using a multimodal framework. Our approach to quantifying attention is grounded in the attention model proposed by [27] and supported by factor analysis of psychometric assessments, incorporating internal factors for a comprehensive evaluation of attention. This approach integrates neuropsychological tests, which serve as the ground truth for attention assessment, with physiological signals such as heart rate variability (HRV) and galvanic skin response (GSR), as well as non-intrusive behavioural cues, including eye tracking and webcam-based monitoring, to develop a comprehensive understanding of attention dynamics. Webcam-based monitoring captures observable indicators such as gaze patterns, facial expressions, and posture, which correlate with attention. These multimodal measures are cross-referenced against neuropsychological test results to ensure reliability and clinical relevance in identifying ADHD-related attention challenges.

Additionally, the study investigates how attention varies across different sensory modalities during digital learning and explores the interplay between cognitive, behavioural, and physiological data. Using AI-driven analysis, the study aims to provide real-time, adaptive interventions tailored to individual needs, reducing reliance on subjective reporting. This approach emphasizes targeted, actionable insights that address specific attention challenges, paving the way for personalized and effective digital learning strategies.

By focusing on real-time assessment and intervention, this research seeks to address the unique challenges faced by children with ADHD during digital learning, offering pathways for more effective and tailored digital learning experiences. The findings will contribute to advancing educational technology for neurodevelopmental disorders and highlight the potential of AI in improving learning environments for children with ADHD.

2 Related Work and Motivation

Previous research on ADHD and digital learning has focused on single aspects of attention. Eye movement was used in the study by the author [8, 23] to investigate behavioural changes in ADHD participants to classify ADHD or non-ADHD. In another study by the author [2, 24], attention levels in children with ADHD were assessed using a video game and brain activity was measured with a single-channel brain-computer interface headset to perform classification of ADHD children. Another study classified ADHD using brain waves and corporal posture [3, 11], uses thermal image and Internet of Things [16] while other uses EEG [1] to detect attention. However, ADHD students exhibit diverse and atypical behaviours due to their unique learning differences [7]. In [25], the author proposes a gaze-based visual attentive interface with adaptive colour schemes to extend the attention span of children with ADHD during reading tasks. In [28], the author highlights the need for a multi-stakeholder approach in designing for children with ADHD, emphasizing the importance of communication and reflection to enhance their well-being and involving relevant stakeholders beyond just a medical model of disability.

To my knowledge, there has been no research aimed at assessing the attention of students with ADHD in real-time while providing immediate feedback. Also, relying on a single behavioural pattern to generalize their attention would be misleading. A thorough examination of diverse behavioral patterns—including types of attention, impulsivity, hyperactivity, and other relevant behaviors—offers a more comprehensive understanding of an individual's attention profile. Furthermore, traditional tests may not serve as a valid ground truth for ADHD, as they often fail to capture the distinctive cognitive and behavioural traits associated with this disorder, such as attention deficits, impulsivity, hyperactivity, and atypical learning styles. Neuropsychological assessments are tailored to evaluate these specific areas, making them more appropriate for individuals with ADHD. Additionally, the integration of physiological, non-intrusive behavioural cues, and neuropsychological measures in the assessment of attention deficits can provide objective and quantifiable data. This approach minimizes subjective biases and enhances the accuracy and validity of the assessment process for students with ADHD.

3 Research Questions

The primary aim of my dissertation is to investigate and develop innovative methods that analyze attention among children with ADHD in digital learning environments, ultimately improving their learning experiences through targeted interventions. This central goal is broken down into four key research questions (RQs):

- **RQ1: Can neuropsychological tests and webcam-based behavioural cues accurately predict attention levels in children during digital learning?**
  This question seeks to explore the effectiveness of combining traditional neuropsychological tests, which are widely used for assessing cognitive functions, with modern technology, such as webcam-based monitoring, to provide a more comprehensive understanding of attention levels in real-time. By examining the correlation between test results and observed behaviours during digital learning tasks, this research aims to identify reliable predictors of attention that can inform tailored educational strategies.
- **RQ2: How do physiological and neuropsychological measures help understand attention deficits in the ADHD population compared to neurotypical students?**
  This question focuses on the comparative analysis of physiological responses (HRV and GSR) and behavioural cues (eye tracking and facial behaviour) alongside neuropsychological measures to elucidate the specific characteristics of attention deficits in children with ADHD compared to neurotypical students. Understanding these differences will enhance our grasp of how ADHD affects learning and attention, thereby

providing a basis for targeted interventions that address the unique needs of this population.

- **RQ3: What are the variations in attention allocation across modalities (e.g., visual, auditory) in children with ADHD during digital learning activities?** This question investigates how children with ADHD distribute their attention across different sensory modalities while engaged in digital learning tasks. By identifying patterns in attention allocation—such as which modalities elicit stronger engagement or distraction—this research aims to inform the development of adaptive content delivery systems that align with the strengths and challenges of ADHD learners, thereby enhancing attention and learning outcomes.
- **RQ4: How can neuropsychological tests and non-intrusive behavioural cues be integrated into AI-based programs for diagnosing and monitoring attention in children with ADHD during digital learning and facilitating timely interventions to improve their attention skills?** This question addresses the application of research insights by integrating multimodal data into AI-driven systems. The goal is to develop systems that can not only diagnose attention in real-time but also provide personalized interventions based on continuous monitoring of a child's attention patterns. This approach can potentially create adaptive learning environments that proactively support children with ADHD in their educational journeys.

Figure 1 illustrates the thesis framework that integrates multimodal data collection, attention analysis, and real-time interventions to enhance cognitive engagement in ADHD digital learning environments. The pipeline combines physiological signals, behavioural cues, and neuropsychological tests to track attention dynamics and provide adaptive feedback to learners.

4 Results and Contribution To Date

4.1 Investigating Neuropsychological Tests and Behavioral Cues for Attention Assessment in Digital Learning

Objective: To investigate the efficacy of neuropsychological tests in detecting attention levels during cognitive engagement tasks in a digital learning environment and to propose and validate the Nuanced Attention Labeling method as an effective approach for capturing subtle changes in attention during digital learning activities, thereby enhancing our understanding of attention dynamics in such contexts.

Research: This work [32, 35] was conducted to address RQ1 and provide insights into RQ2. I used a clinical-based attention model to assess various components of attention relevant to digital learning environments. Although previous studies have explored methods for assessing student attention in digital learning environments, these approaches often show inconsistencies in linking engagement and attention, raising doubts about their accuracy [12, 14, 19, 29, 31]. Sometimes, they rely on explicit ground truth collection, which may introduce cognitive load and distractions. To overcome this limitation, I utilized standardized neuropsychological assessments, which are crafted to evaluate various facets of attention, including selective, sustained, and alternating attention among participants. Specifically, I used the Stroop Test for selective attention, the Continuous Performance Test (CPT) for sustained attention, and the Trail Making Test (TMT) for alternating attention, highlighting their relevance in pedagogical contexts.

For this research, I collected data from 98 participants, including webcam recordings and cognitive task responses in CSV format, facilitating the computerized administration of various neuropsychological tasks. I hypothesized that their performance and reaction times could serve as labels for attention. Since this is a novel approach, I adopted a comparative method involving external annotators to support my hypothesis by establishing convergent validity. The strong correlation between the annotated facial video data and neuropsychological test scores reinforces the validity of using these tests in educational settings. Additionally, my aim was to highlight how fine-grained labelling enables a detailed examination of cognitive engagement, anticipating reduced errors in capturing subtle variations in attention. These findings suggest that neuropsychological tests can effectively measure attention levels, providing a more nuanced and reliable means of assessing student attention compared to traditional self-reporting methods in a digital learning environment.

4.2 Enhancing Inclusive Education Through Computational Frameworks for Textbook Adaptation

Objective: To foster inclusive education by developing a computational framework to adapt textbooks for inclusive education, removing decorative images to improve clarity and reduce distractions for students with ADHD.

Research: This work [33] addressed aspects of RQ4, focusing on intervention strategies deployed when a student is inattentive, with inattention identified through neuropsychological tests and behavioural cues. While full integration into AI-based programs is ongoing, this research underscores the potential of targeted interventions to enhance attention in children with ADHD. Recognizing that students with ADHD are easily distracted by colours and images in textbooks can lead to more effective learning strategies [21, 30], I propose a computational framework that adapts educational materials to their needs. This framework categorizes images into *Essential, Informative,* and *Decorative* classes with expert help. It helps adapt textbooks by customizing image inclusion based on user needs and improving clarity by removing or altering decorative images. Although automating image classification poses challenges due to content variability, inconsistent image quality, and the need for expert annotation, it offers a scalable solution for creating inclusive educational materials.

I extracted images from three textbooks in PDF format and grouped the content into activity blocks based on layout, font style, and spacing. Images were linked to blocks according to their page position. The variability in textbook layouts posed challenges for standardization, especially in adapting materials for students with ADHD, who are prone to distraction. Experts manually annotated the images into *Essential, Informative,* and *Decorative* categories, though distinguishing between *Essential* and *Informative* proved

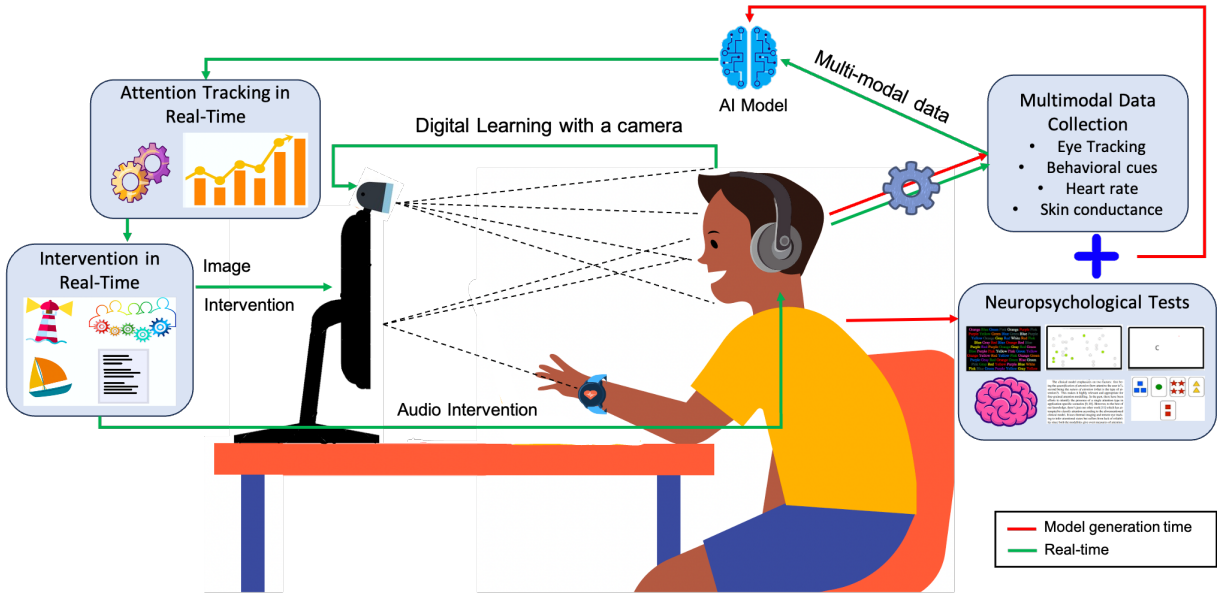


Figure 1: Thesis framework for attention analysis and real-time intervention in ADHD digital learning, using multimodal data and neuropsychological tests to enhance cognitive engagement.

challenging due to subtle differences. I developed a framework using multimodal and monomodal approaches with Machine Learning (ML) models, identifying specific text-image patterns to categorize images. This process helps retain educational content while automatically removing non-essential images, reducing distractions for ADHD students and supporting a more focused learning experience.

5 Future Work and Challenges

In the future, I plan to expand this research by addressing RQ2 and RQ3 through collaborations with specialized schools for children with ADHD. These collaborations will collect multimodal data in real-world educational settings, allowing a deeper understanding of ADHD-specific attention patterns. However, one significant challenge will be gaining access to these schools and obtaining consent from parents and guardians, which is essential for ethical research practices. Additionally, collecting data in a naturalistic environment may present logistical difficulties, such as coordinating schedules with educators and managing the variability in classroom dynamics.

This research will compare attention dynamics between ADHD and neurotypical students to identify specific differences and similarities in their attention in digital learning environments. Understanding these patterns will allow for the refinement of interventions aimed at improving attention in ADHD populations.

To improve measurement accuracy, I aim to integrate physiological and behavioural data into ML models, correlating these inputs with established neuropsychological tests to ensure validity. By capturing a broader range of cues, such as eye-tracking, pupil dilation, facial expressions, GSR and heart rate variability, the model can become more sensitive to fluctuations in attention across different modalities. However, challenges may arise in ensuring the reliability of the data collected from various sensors, as well as the potential need for extensive pre-processing to handle noise and variability in the data.

Another aspect of future work will be evaluating multimodal versus mono-modal approaches in assessing attention, also focusing on how different sensory inputs (e.g., auditory, visual) influence attention. This will be critical for designing adaptive digital learning interfaces that adjust content delivery in real-time based on each student's attention level. Nevertheless, evaluating these approaches may involve challenges related to the integration of diverse data sources and the complexity of analyzing multimodal data sets.

Finally, I will validate these adaptive interventions in classroom settings to assess their long-term impact on attention and learning outcomes. Collaboration with educators will involve workshops to tailor interventions, while psychologists will provide insights into developmental appropriateness and ethical considerations. However, ongoing challenges may include obtaining sufficient long-term data to draw meaningful conclusions about the efficacy of these interventions and addressing potential resistance from educational institutions in implementing new assessment and intervention strategies. The outcomes of this research will support the development of evidence-based adaptive digital learning tools that educators can easily implement, reducing reliance on extensive technical expertise.

6 Contributions

When addressing the research questions, the following contributions are expected:

- Enhanced Understanding of ADHD Attention (RQ1, RQ2): This work will provide insights into how physiological signals and behavioural cues can be used to assess attention in children with ADHD and how their attention deficits differ from neurotypical students in digital learning environments.

- Exploration of Multimodal Attention (RQ3): The study will explore attention shifts across sensory channels, helping to develop targeted interventions for ADHD students during digital learning.
- AI-Driven Attention Interventions (RQ4): By integrating neuropsychological assessments and behavioural cues into AI systems, this research aims to develop real-time interventions to help ADHD students maintain focus.
- Personalized Educational Resources: A framework for classifying textbook images will enable tailored content delivery, reducing distractions and improving attention for ADHD students.

These contributions will significantly improve the digital learning experience for ADHD students by fostering more personalized, inclusive, and adaptive educational environments, which can lead to better focus, engagement, and academic success.

7 Research Status

I am a third-year PhD candidate from the Human Machine Interaction (HMI) Lab at IIIT-Delhi, working under the supervision of Dr Jainendra Shukla. I have successfully completed my comprehensive exam (PhD confirmation milestone) in May 2024, and I expect to submit my PhD thesis in December 2025. Indian PhD programs usually span 4-5 years. I attended another doctoral consortium at the AI in Education Conference 2023 [34]. While that experience was valuable, I am particularly eager to gain insights and input from the CHI community, which encompasses a broader domain of Human-Computer Interaction. Attending CHI will give me a wider perspective on interdisciplinary approaches and methodologies, enhancing my research in a unique way to this prestigious conference.

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